Lab 1: Symmetry in Deep Learning Representations

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Question 1: Optimal Parameters for Sum Aggregation

Problem Statement: We aim to determine the optimal parameters for a DeepSets architecture defined by $f(X) = \rho(\sum_{x \in X} \phi(x))$ to learn the sum of a set of integers $x \in \{1, \ldots, 10\}$.

Theoretical Derivation

To solve this task, the model must approximate the identity function. Assuming the first fully connected layer uses a ReLU activation function ($ReLU(z) = \max(0, z)$), the derivation is as follows:

1. **Representation Network (ϕ):** The network must pass the input values through to the aggregator without distortion. Ideally, $\phi(x) = x$. For a fully connected layer $z = W_1 x + b_1$:
   - We set weights $W_1 = 1$ and bias $b_1 = 0$.
   - Since inputs are strictly positive integers ($x \geq 1$), the ReLU activation is in its linear region:
   $$ReLU(1 \cdot x + 0) = x \quad \forall x \in \{1, \ldots, 10\}$$
2. **Regressor Network (ρ):** The aggregator sums the outputs of ϕ, yielding $S = \sum \phi(x) = \sum x$. The final network must output this sum directly. For the final layer $y = W_2 S + b_2$:
   - We set weights $W_2 = 1$ and bias $b_2 = 0$.
   - This yields $y = 1 \cdot S + 0 = \sum x$.

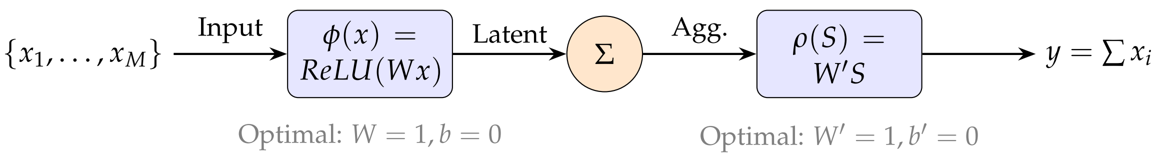

Figure 1: Architecture flow for the Sum Task. The network learns Identity mappings.

Conclusion Q1

The DeepSets architecture is expected to learn **weights equal to 1** and **biases equal to 0** for both ϕ and ρ to perform an exact sum aggregation on positive integers.

Question 2: Distinguishing Sets with Zero Sum

Analysis: Yes, the DeepSets model can learn different representations for the four sets, even though the sum of the raw elements for each set is equal to $[0, 0]$.

The Role of Non-Linearity

The core principle of DeepSets is the decomposition $f(X) = \rho(\sum_{x \in X} \phi(x))$. The aggregation is applied to the *latent representations* $\phi(x)$, not the raw inputs x.

- **Linear Case (Failure):** If ϕ is linear (e.g., $\phi(x) = Ax$), then $\sum \phi(x) = A \sum x = A \cdot 0 = 0$. All sets would map to zero and be indistinguishable.
- **Non-Linear Case (Success):** With a non-linear MLP, ϕ breaks the symmetry. For example, if the model learns a quadratic function approximating the squared L2 norm $\phi(x) = \|x\|^2$, the sets become separable.

Visualization of Separation

Consider Set 1 and Set 2 from the provided dataset. We demonstrate how a quadratic ϕ separates them:

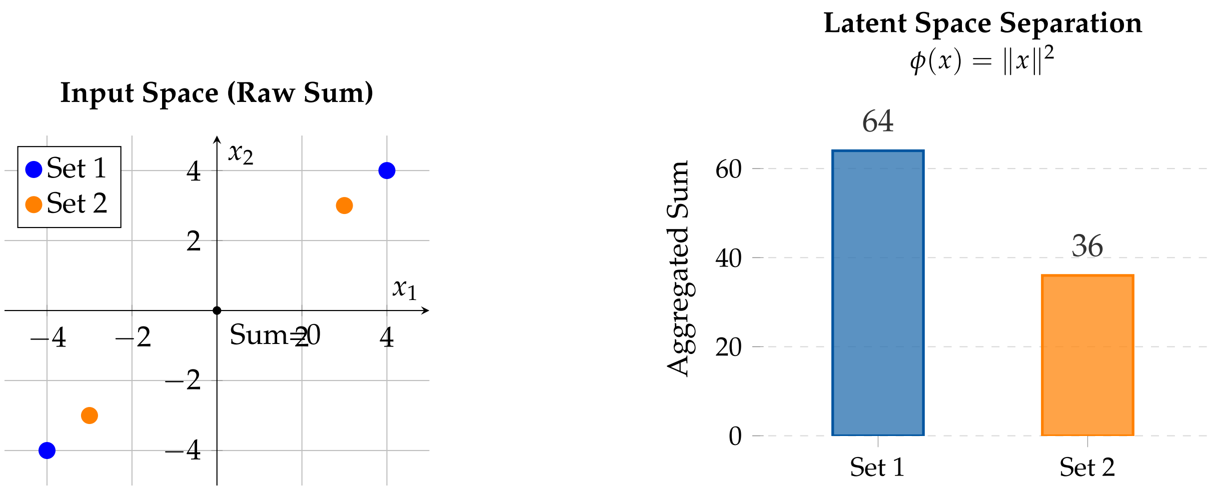

Figure 2: Left: In input space, both sets sum to $(0, 0)$. Right: In latent space (using a quadratic ϕ), the aggregated sums are 64 and 36, making them easily separable by ρ.

Since the aggregated sums of the representations are distinct ($64 \neq 36$), the final network ρ can distinguish the sets and map them to distinct outputs.

Annexe

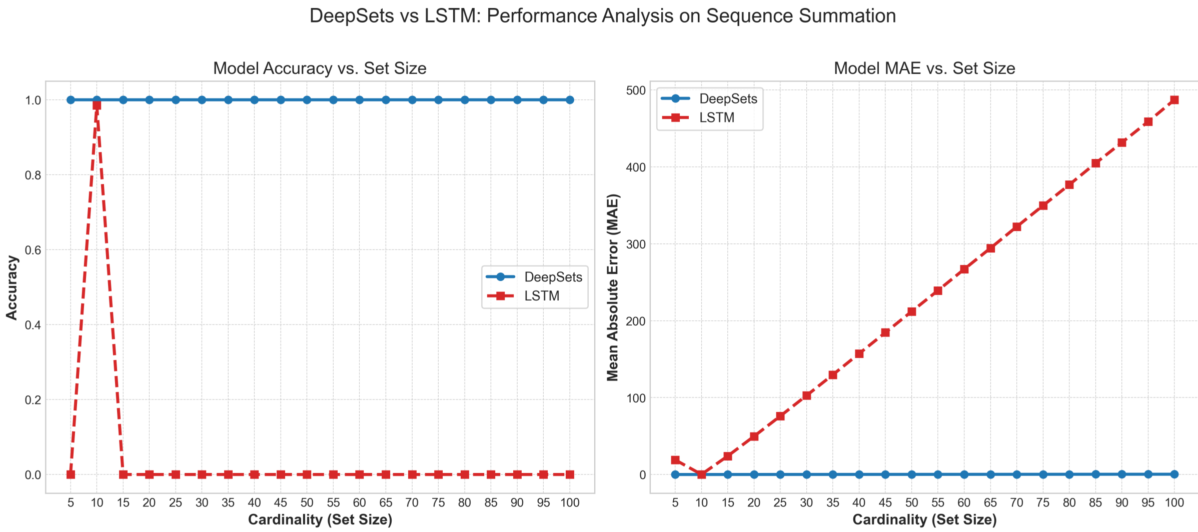

Figure 3: Caption